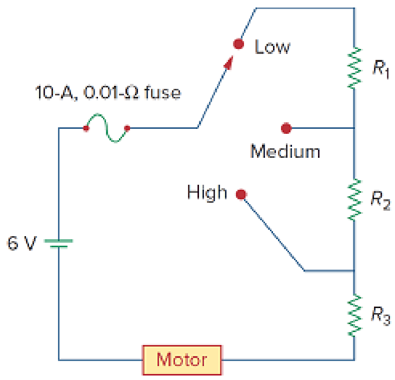
Problem

The circuit in Fig. 2.134 is to control the speed of a motor such that the motor draws currents 5 A, 3 A, and 1 A when the switch is at high, medium, and low positions, respectively. The motor can be modeled as a load resistance of 20 mΩ. Determine the series dropping resistances *R*1, *R*2, and *R*3.

Figure 2.134:



Step-by-step solution

Step 1 of 8

Consider the following circuit diagram:

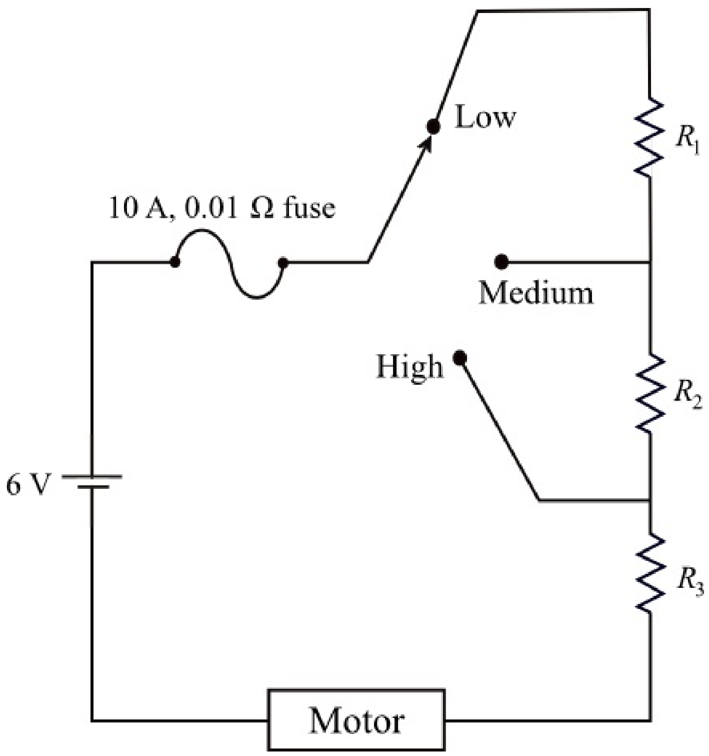


Figure 1

Step 2 of 8

Consider the following data:

Current drawn by the motor when switch is at high position, $I_{\text{high}} = 5\text{ A}$

Current drawn by the motor when switch is at medium position, $I_{medium} = 3\text{ A}$

Current drawn by the motor when switch is at low position, $I_{low} = 1\text{ A}$

Load resistance of the motor, $R_L = 20\text{ m}\Omega$

Fuse resistance, $R_{fuse} = 0.01\ \Omega$

Step 3 of 8

Draw the circuit diagram, when switch is at high position.

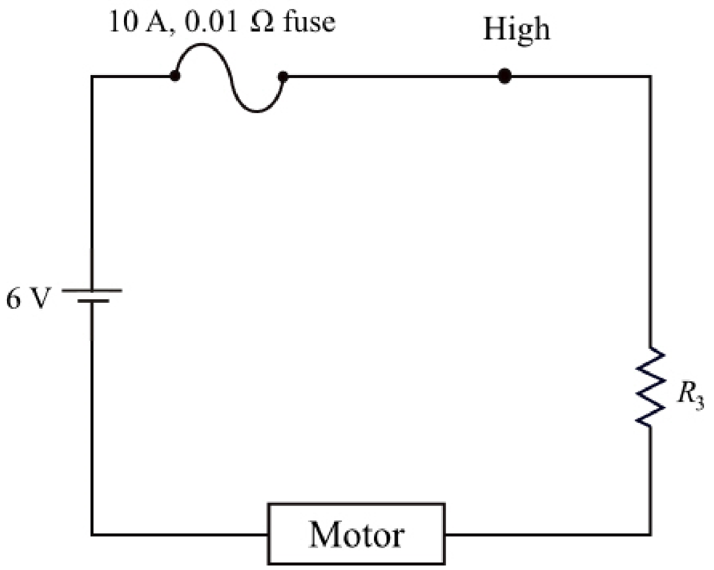


Figure 2

Step 4 of 8

Apply Kirchhoff's voltage law to loop in Figure 2.

$$-6 + \left(0.01 + R_3 + 20\times10^{-3}\right) I_{high} = 0$$
$$(0.01 + R_3 + 0.02)(5) = 6$$
$$(0.03 + R_3)(5) = 6$$
$$0.03 + R_3 = 1.2$$

$$R_3 = 1.17\ \Omega$$

Therefore, the series dropping resistance, R_3 is $\boxed{1.17\ \Omega}$.

Step 5 of 8

Draw the circuit diagram, when switch is at medium position.

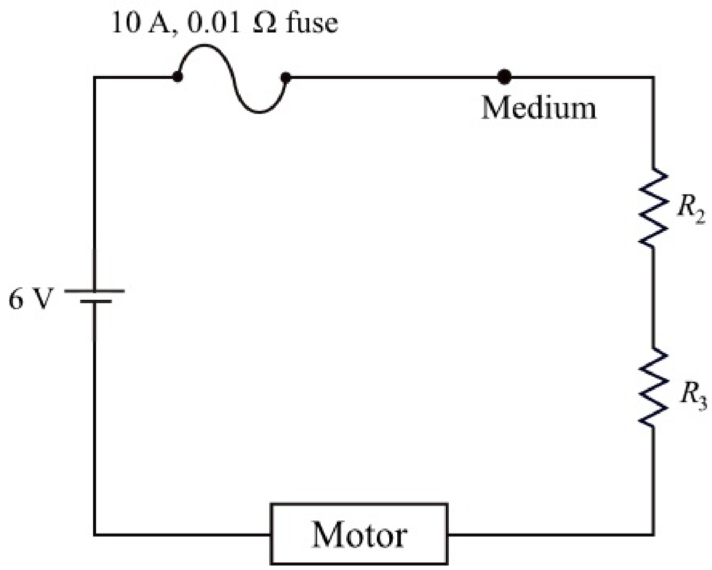


Figure 3

Step 6 of 8

Apply Kirchhoff's voltage law to loop in Figure 3.

$$-6+\left(0.01+R_3+R_2+20\times10^{-3}\right)I_{medium}=0$$
$$(0.01+1.17+R_2+0.02)(3)=6$$
$$(1.2+R_2)(3)=6$$
$$1.2+R_2=2$$
$$R_2=0.8\ \Omega$$

Therefore, the series dropping resistance, R_2 is $\boxed{0.8\ \Omega}$.

Step 7 of 8

Draw the circuit diagram, when switch is at low position.

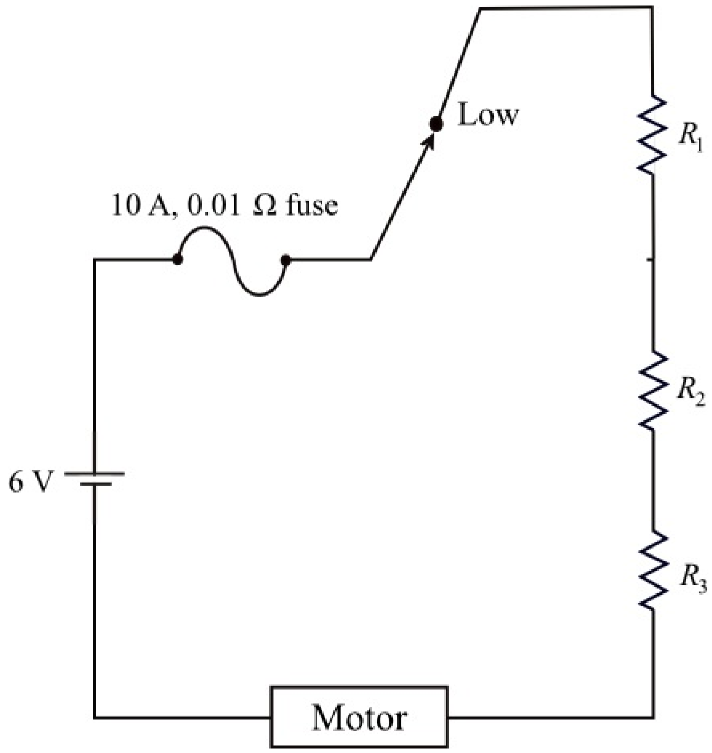


Figure 4

Step 8 of 8

Apply Kirchhoff's voltage law to loop in Figure 4.

$$-6+\left(0.01+R_3+R_2+R_1+20\times10^{-3}\right)I_{low}=0$$
$$(0.01+1.17+0.8+R_1+0.02)(1)=6$$
$$(2+R_1)(1)=6$$
$$2+R_1=6$$
$$R_1=4\ \Omega$$

Therefore, the series dropping resistance, R_1 is $\boxed{4\ \Omega}$.